AWS Cost Analysis

University Management System

Infrastructure cost baseline, seasonal enrollment scaling, and savings roadmap

Baseline Cost

Normal operating days - 356 days / year

Component	Specs	Qty	Hourly Rate	Est. Monthly Cost
Web EC2	`t3.small (Ubuntu)`	1	$0.024	~$17.52
PgBouncer EC2	`t3.small (Ubuntu) - centralized`	1	$0.024	~$17.52
Database	`RDS PostgreSQL db.t3.medium`	1	$0.068	~$49.64
Storage	`EBS gp3 (~110GB total)`	-	$0.095/GB	~$10.45
Total Baseline				**~$95.13 / month**

Pricing basis: AWS on-demand hourly rates, us-east-1, 730 hrs/month average. Costs shown are pre-discount (on-demand) baseline.

Enrollment Burst Cost

9 days/year - 3 enrollment periods - impact is negligible

Component	Specs	Qty	Hourly Rate	9-Day Cost
Application Load Balancer	`ALB - enrollment burst only`	1	$0.0225	~$4.86
Auto Scaling Group EC2	`t3.small (Ubuntu) x5`	5	$0.024	~$25.92
Total Enrollment Add-on				**~$30.78 / year**

Deployment automation - decision pending: `Lambda + EventBridge` (event-driven) vs. `Terraform` (IaC) to trigger the ALB and the existing Auto Scaling Group, which already spins up 5 EC2 instances during enrollment.
Impact: the added cost of these 9 days is negligible against the annual baseline.

Cost Optimization Roadmap

Commitment-based pricing can cut the monthly baseline by roughly half

AWS recommends Compute Savings Plans over Reserved Instances for EC2 - same discount depth, plus flexibility to shift instance family, size and OS.

Resource	Current Model	Recommended Option	Term	Est. Savings
EC2 Compute (Web + PgBouncer)	On-Demand	Compute Savings Plan	1-yr / 3-yr	30% - 55%
RDS PostgreSQL	On-Demand	Reserved Instance	1-yr / 3-yr	up to 60%

Projected monthly impact

Resource	Baseline / mo	Optimized Model	Optimized / mo
EC2 Compute	$35.04	Savings Plan (30-55%)	$15.77 - $24.53
RDS PostgreSQL	$49.64	Reserved Instance (~60%)	~$19.86
Storage (EBS)	$10.45	No change	$10.45
Total	**$95.13**		**~$46 - $55**

Summary & Recommendation

~52%

potential monthly cost reduction with 3-year Savings Plan + RDS Reserved Instance commitments

- **Compute Savings Plan for EC2:** cheaper and more flexible than Reserved Instances for the Web and PgBouncer instances.
- **Reserved Instance for RDS:** 1 or 3-year term cuts the database bill by about 60%.
- **Enrollment burst is negligible:** 9 days/year of ALB + Auto Scaling Group adds only ~$30.78/year.
- **Automation decision pending:** Lambda + EventBridge vs. Terraform to trigger the ALB deployment.
- **Architecture diagram:** full draw.io diagram to be attached on the architecture slide.

Next steps

Approve Savings Plan and Reserved Instance terms, finalize the ALB automation approach, and attach the draw.io architecture link before the next enrollment cycle.